

Name: _____ Class: _____ Date: _____

Universal STEM Worksheet Engine

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AI Agent Competition

Secondary STEM Category | ISNSW Curriculum Conference 2026

Primary → Stage 6 · Pure TikZ, zero embedded images · NESA-aligned

One engine, one design language — from a Year 2 child measuring a friendly stick-figure with a ruler, to a Stage 6 student resolving forces, analysing AC circuits and proving roots of unity. Every diagram is generated in **pure TikZ**: no embedded images. Outcome codes are **indicative drafts for teacher verification** against the current NSW NESA syllabuses. **Note:** Document layout is highly compressed for ISNSW showcase purposes. In-production artifacts generate with stage-appropriate spacing to manage cognitive load.

■ Primary · K–6 ■ Junior STEM · Yr 7–10 ■ Senior STEM · Yr 11–12 ■ Extension

PRIMARY Early Years · Time & Measuring Length

Learning Intentions & Success Criteria

I can read analogue time, measure length in centimetres starting from 0, and compare heights using a scale.

Telling the Time

Y2 Analogue clock [2 marks]

NESA: Measurement & Geometry

The clock shows a time in the morning. Write it in 12-hour and 24-hour formats.



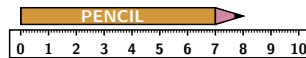
12hr: _____ 24hr: _____

Using a Ruler

Y2 Measure pencil [1 mark]

NESA: Measurement & Geometry

Read the ruler and state the exact length of the pencil in centimetres.

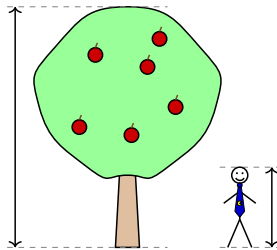


Length: _____ cm

Comparing Heights

Y2 How tall? [2 marks]

NESA: Measurement & Geometry

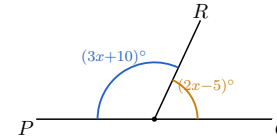
Use the scale to measure the **height** of the tree and Darra. How much taller is the tree?
Tree: _____ cm
Darra: _____ cm
Difference: _____ cm

JUNIOR STEM Stage 4–5 · Geometry, Algebra, Trigonometry & Science

Angle Reasoning

Stage 4 Angles on a line [2 marks]

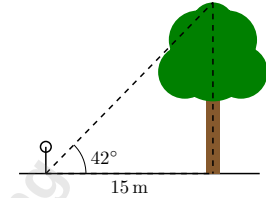
NESA: Angle relationships

Line POQ is straight and ray OR meets it at O . Form an equation, solve for x , then state $\angle POR$.

Angle of Elevation

Stage 5 Trigonometry [2 marks]

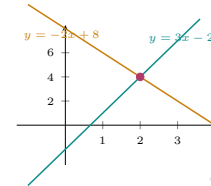
NESA: Trigonometry & Pythagoras

Standing 15 m from a tree, a clinometer reads 42° to the top. Find the height to 2 d.p.

Simultaneous Lines

Stage 4 Point of intersection [2 marks]

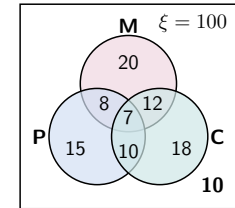
NESA: Linear relationships

The lines $y = 3x - 2$ and $y = -2x + 8$ are graphed below. State their point of intersection.

Set Theory & Probability

Stage 5 Venn diagrams [3 marks]

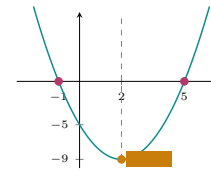
NESA: Probability / data

100 students study Maths (M), Physics (P) or Chemistry (C). Find $P(P | M)$.

The Parabola

Stage 5 Quadratic features [3 marks]

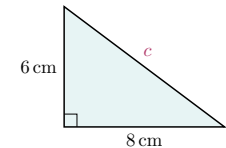
NESA: Non-linear relationships

For $y = x^2 - 4x - 5$, state the x -intercepts, axis of symmetry, and the turning point.

Pythagoras

Stage 5 Find the hypotenuse [2 marks]

NESA: Trigonometry & Pythagoras

A right-angled triangle has the two shorter sides shown. Find the length of the hypotenuse c .

$$c = \sqrt{a^2 + b^2}$$

Investigation Skills

Stage 5 Science — Pendulum Investigation

[4 marks]

A simple pendulum is released and you investigate how **string length** affects the **period** of one swing. State the independent and dependent variables, **two** controlled variables, a testable hypothesis, and **one** safety risk with its control. *NESA: Working Scientifically — questioning, predicting, planning*

IV: _____

Hypothesis / risk: _____

DV: _____

Controlled: _____, _____

SENIOR STEM Stage 6 · Physics, Chemistry, Software & Extension

Learning Intentions & Success Criteria

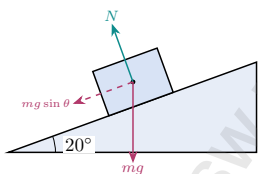
I can resolve forces on an incline, analyse a series RLC circuit at resonance, operate with 3D vectors, represent organic reactions, prove results about roots of unity, interpret a slope field, model a normal distribution, and reduce a logic circuit.

Mechanics · Inclined Plane

Physics Free-body diagram [3 marks]

NESA: Physics — Dynamics

A block of mass $m = 6.0 \text{ kg}$ rests on a **frictionless** incline at $\theta = 20^\circ$. Draw the free-body diagram and find the acceleration.

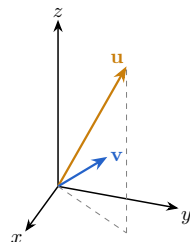


Extension · 3D Vectors

Maths Ext Dot product & angle [3 marks]

NESA: Enrichment — vectors in 3D

For $\mathbf{u} = 2\mathbf{i} + 3\mathbf{j} + 6\mathbf{k}$ and $\mathbf{v} = \mathbf{i} + 2\mathbf{j} + 2\mathbf{k}$, find $\mathbf{u} \cdot \mathbf{v}$ and the angle between them.

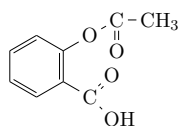


Chemistry · Organic

Chem Functional groups & synthesis [4 marks]

NESA: Chemistry — Organic Chemistry

Below is the skeletal structure of Aspirin. Identify the three primary functional groups present. Write the balanced chemical equation for the synthesis of Aspirin reacting salicylic acid with acetic anhydride.



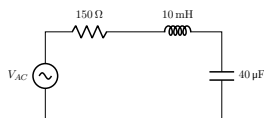
Equation: _____

Electromagnetism · RLC

Physics Series resonance [3 marks]

NESA: Physics — Electromagnetism

For the series RLC circuit with $R = 150 \Omega$, $L = 10 \text{ mH}$ and $C = 40 \mu\text{F}$, calculate the resonant frequency in hertz.



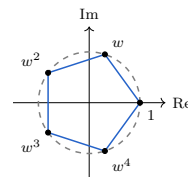
$$f_r = \frac{1}{2\pi\sqrt{LC}}$$

Extension · Complex Numbers

Ext 2 Roots of unity [3 marks]

NESA: Mathematics Extension 2

The diagram shows the 5th roots of unity, where $w = \text{cis}\left(\frac{2\pi}{5}\right)$. Prove $1 + w + w^2 + w^3 + w^4 = 0$.

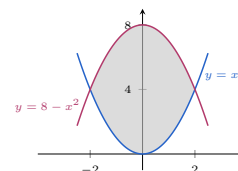


Calculus · Area Between Curves

Maths Adv Definite integral [3 marks]

NESA: Mathematics Advanced

Find the exact area of the shaded region bounded by $y = x^2$ and $y = 8 - x^2$.

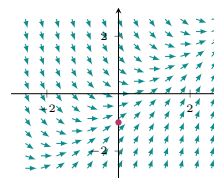


Calculus · Slope Field

Maths Ext Direction field [3 marks]

NESA: Mathematics Extension 1 — differential equations

The slope field for $\frac{dy}{dx} = x - y$ is shown. Sketch the solution curve passing through $(0, -1)$ and describe its long-term behaviour.

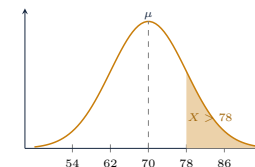


Statistics · Normal Distribution

Maths Adv The empirical rule [2 marks]

NESA: Mathematics Advanced — statistical analysis

Test scores are normally distributed with mean $\mu = 70$ and standard deviation $\sigma = 8$. Use the shaded region to estimate $P(X > 78)$.



Multiple Choice · Logarithms

Maths Adv Logarithmic equations [1 mark]

NESA: Mathematics Advanced

What is the solution to $\log_2(x) + \log_2(x - 3) = 2$?

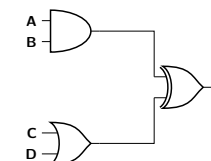
- A. $x = -1$
- B. $x = 4$
- C. $x = 4$ and $x = -1$
- D. No real solutions

Software & Engineering

Soft Design Boolean circuit [3 marks]

NESA: Software Engineering

Write the Boolean expression for the output Y , then state the value of Y when $A = 1$, $B = 1$, $C = 0$, $D = 0$.

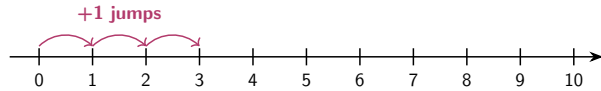


SHOWCASE Scalable Visual Pedagogy & Cross-Curricular**Learning Intentions & Success Criteria**

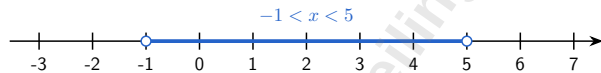
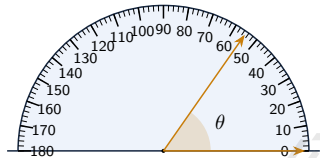
Competition Note: The following components demonstrate the engine's zero-shot ability to scale a single visual framework across distinct educational stages (Year 2 vs. Stage 6) and adapt to entirely different faculties (History, English) using pure TikZ.

Component 1 · The Number Line — One Macro, Any Stage

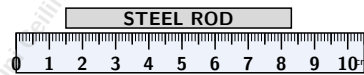
Year 2: Count the jumps to find where the kangaroo lands.



Stage 6: The *same* coordinate axis component now represents the solution of $|x - 2| < 3$.

**Component 2 · Visual Protractor**

Measure θ to the nearest degree.

Component 3 · Precision Ruler

Read the rod's length to the nearest millimetre.

Cross-Curricular · Historical Timeline**Modern History** Timeline Analysis

NESA: Modern History

The timeline below maps four critical flashpoints of the Cold War. Select ONE of these events and evaluate its long-term impact on United States-Soviet Union geopolitical relations.



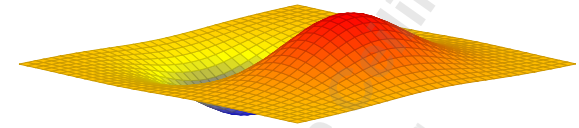
Cross-Curricular · Literary Analysis**English Adv** Translation & Literary Devices**[4 marks]**

NESA: English Advanced

Translate the following excerpt from Shakespeare's *Hamlet* into modern Australian vernacular. Additionally, identify and analyse the primary literary device used in the final two lines to convey mental anguish.

William Shakespeare, Hamlet

To be, or not to be, that is the question:
Whether 'tis nobler in the mind to suffer
The slings and arrows of outrageous fortune,
Or to take arms against a sea of troubles...

Component 4 · Advanced 3D Topology & Data Vis

Advanced PGFPlots: Render complex 3D topological surfaces and multivariable calculus functions natively without relying on external image rendering engines.

Cross-Curricular · Business Studies**Business** Strategic Management**[3 marks]**

NESA: Business Studies

A legacy automotive company has invested \$500 million into developing a highly efficient internal combustion engine. Before completion, a breakthrough in solid-state batteries makes electric vehicles significantly cheaper to produce. The board argues they must launch the new combustion engine to "recover our investment." Identify the logical fallacy driving the board's argument and explain why proceeding could harm their long-term market position.

AI Pedagogy Engines · Deep Learning Extensions**Spot the Error** Calculus bounds**[2 marks]**

A student evaluates $\int_0^2 2x\sqrt{x^2+1} dx$.

L1: Let $u = x^2 + 1$, $du = 2x dx$.

L2: $= \int_0^2 \sqrt{u} du$

L3: $= \left[\frac{2}{3} u^{3/2} \right]_0^2 = \frac{4\sqrt{2}}{3}$.

Error in Line: _____

Parameter Shift Dynamics**[2 marks]**

Refer to the **Inclined Plane** question. If the angle θ is increased to 30° and the mass doubled to 12.0 kg :

a) Does acceleration increase, decrease, or stay the same?

b) Does the normal force N increase or decrease?

ASSESS Automated Marking System & Feedback

TikZ-Typeset Marking Rubric — Working Mathematically / Scientifically

Criterion	Developing (1)	Proficient (2)	Exemplary (3)	Mark
Method	Partially valid approach	Selects a correct method	Efficient, fully justified method	
Working	Steps unclear or gaps	Logical steps, minor slips	Complete, clearly sequenced	
Accuracy & units	Frequent errors; units missing	Mostly accurate; correct units	Accurate; units & sig figs correct	
Communication	Hard to follow	Clear notation	Concise, syllabus-aligned notation	

Mark Allocation & Syllabus Map

Band	Focus areas sampled	Marks
Primary (K–6)	Time, length, comparing height	5
Junior STEM (Yr 7–10)	Angles, linear, quadratic, trig, probability, Pythagoras, investigation	18
Senior STEM (Yr 11–12)	Dynamics, electromagnetism, organic chemistry, calculus, slope fields, statistics, logic	22
Extension	3D vectors, roots of unity	6
Total: _____		51

What Went Well

Even Better If

Overall: ☐ Developing ☐ Proficient ☐ Exemplary

Grade: _____

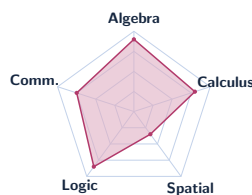
AI Tutor Diagnostic · Cognitive Profile & Adaptive Remediation

Algorithmic Insight (Mental Maths Mentor):

Student demonstrates high procedural fluency in algebraic manipulation and calculus, but exhibits cognitive overload when translating text to visual-spatial representations (e.g., 3D Vectors, Geometry).

Adaptive Next Steps:

- **Decrease:** Repetitive procedural algebra tasks.
- **Increase:** GeoGebra-linked spatial reasoning tasks.
- **Mentorship Note:** High resilience shown in multi-step problem solving. Acknowledge effort and pivot to visual conceptual scaffolding for the next module.



ANSWERS Teacher Answer Key — Worked Reasoning

Time. 12-hour: 10:10 AM; 24-hour: 10:10.

Ruler. 7.0 cm (the pencil tip aligns with the 7 mark; insist the student starts at 0).

Heights. Read each double-arrow scale: tree ≈ 6 units, Darra ≈ 2 units; the tree is about 4 units taller (accept consistent measured values).

Angles. POQ straight $\Rightarrow \angle POR + \angle ROQ = 180^\circ$: $(3x + 10) + (2x - 5) = 180 \Rightarrow 5x + 5 = 180 \Rightarrow x = 35$. Hence $\angle POR = 3(35) + 10 = 115^\circ$.

Lines. $3x - 2 = -2x + 8 \Rightarrow 5x = 10 \Rightarrow x = 2, y = 4$. Intersection (2, 4).

Parabola. $x^2 - 4x - 5 = (x - 5)(x + 1)$; intercepts $x = -1, 5$; axis $x = 2$; turning point (2, -9).

Elevation. $\tan(42^\circ) = \frac{h}{15} \Rightarrow h = 15 \tan(42^\circ) \approx 13.51$ m.

Venn. Only one subject: $20 + 15 + 18 = 53$. $P(P \mid M) = \frac{n(P \cap M)}{n(M)} = \frac{8 + 7}{20 + 8 + 12 + 7} = \frac{15}{47}$.

Pythagoras. $c = \sqrt{8^2 + 6^2} = \sqrt{100} = 10$ cm.

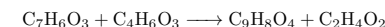
Pendulum. IV: string length; DV: period. Controlled (any two): bob mass, release angle, timing method, location. Hypothesis: longer string $\Rightarrow$ longer period. Risk: bob striking a student — keep the path clear, limit amplitude.

Inclined plane. Frictionless: $a = g \sin \theta = 9.8 \sin(20^\circ) \approx 3.35 \text{ m} \cdot \text{s}^{-2}$ down the plane ($N = mg \cos \theta$ balances the perpendicular component).

RLC. $f_r = \frac{1}{2\pi\sqrt{LC}}$, $L = 10 \times 10^{-3}$, $C = 40 \times 10^{-6}$: $\sqrt{LC} \approx 6.325 \times 10^{-4}$, so $f_r \approx 251.65$ Hz.

3D vectors. $\mathbf{u} \cdot \mathbf{v} = (2)(1) + (3)(2) + (6)(2) = 20$; $|\mathbf{u}| = 7$, $|\mathbf{v}| = 3$, so $\cos \theta = \frac{20}{21} \Rightarrow \theta \approx 18^\circ$.

Chemistry. Functional groups: carboxylic acid, ester, aromatic (benzene) ring. Synthesis:



(salicylic acid + acetic anhydride $\rightarrow$ aspirin + acetic acid).

Roots of unity. The roots solve $z^5 - 1 = 0$. Their sum is the geometric series $\frac{1 - w^5}{1 - w}$; since $w^5 = 1$ the numerator is 0 (and $w \neq 1$), so the sum is 0.

Area between curves. Intersect: $x^2 = 8 - x^2 \Rightarrow x = \pm 2$. $A = \int_{-2}^2 [(8 - x^2) - x^2] dx = \int_{-2}^2 (8 - 2x^2) dx = \frac{64}{3}$ square units.

Slope field. Solution through (0, -1) rises, crosses below the line $y = x - 1$, and is asymptotic to $y = x - 1$ as $x \rightarrow \infty$ (the particular solution is $y = x - 1$).

Normal distribution. $z = \frac{78 - 70}{8} = 1$. By the empirical rule $P(X > 78) = P(Z > 1) \approx 0.16$ (16%).

Logic. $Y = (A \wedge B) \oplus (C \vee D)$. With $A = B = 1$, $C = D = 0$: $(1) \oplus (0) = 1$.

MC Logarithms. B ($x = 4$). $x = -1$ is outside the log domain.

Spot the Error. Line 2. The integral bounds were not changed to u -values (should be $u = 1$ to 5).

Parameter Shift. a) Increases ($a = g \sin \theta$ is mass-independent, and $\sin(30^\circ) > \sin(20^\circ)$). b) Increases (doubling the mass outweighs the slight drop in $\cos \theta$).

Business Studies. Identifies the Sunk Cost Fallacy. Proceeding ignores future viability; EVs will undercut ICE vehicles, losing market share and compounding financial losses.



ISNSW AI Agent Competition



Try the Custom Gem!



Scan to view Agent Instructions

Name: _____

Date: _____

Introduction to Pythagoras' Theorem

Year 8 — Stage 4 Mathematics

Learning Intentions

- I can identify the hypotenuse in a right-angled triangle.
- I can use Pythagoras' theorem to calculate the hypotenuse.
- I can rearrange Pythagoras' theorem to find a shorter side.
- I can apply Pythagoras' theorem to a real-world situation.

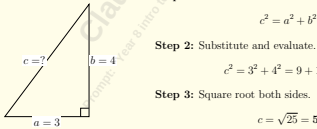
Key Rule

In a right-angled triangle, the **hypotenuse** c (longest side, opposite the right angle) satisfies:

$$c^2 = a^2 + b^2 \implies c = \sqrt{a^2 + b^2}$$

To find a **shorter side**: $a^2 = c^2 - b^2 \implies a = \sqrt{c^2 - b^2}$

Worked Example — Finding the Hypotenuse



Step 1: Write the theorem.

$$c^2 = a^2 + b^2$$

Step 2: Substitute and evaluate.

$$c^2 = 3^2 + 4^2 = 9 + 16 = 25$$

Step 3: Square root both sides.

$$c = \sqrt{25} = 5$$

Name: _____

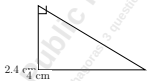
Date: _____

Introduction to Pythagoras' Theorem

Key Idea: In a right-angled triangle, the square of the hypotenuse equals the sum of the squares of the other two sides.

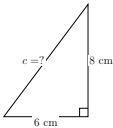
$$c^2 = a^2 + b^2$$

1. A right-angled triangle has legs $a = 3$ cm and $b = 4$ cm. Find the length of the hypotenuse c .



3. A student claims that if $a = 5$ and $b = 12$, then the hypotenuse is $c = 17$ because “you just add the sides”. Explain why this is incorrect and calculate the correct value of c .

1. [Fluency] Find the length of the hypotenuse c in the triangle below. Show all working.



2. [Misconception Check] A right-angled triangle has a hypotenuse of 13 cm and one shorter side of 5 cm.

(a) Find the length of the unknown shorter side a . Show full working.

(b) A student attempted the same problem and wrote:

$$a^2 = 5^2 + 13^2 = 25 + 169 = 194, \quad \text{so} \quad a = \sqrt{194} \approx 13.9 \text{ cm.}$$

Identify the exact error the student made. Then write the correct solution.

Teacher Answer Key

4. Solution:

$$c^2 = 3^2 + 4^2 = 9 + 16 = 25$$

$$c = 5 \text{ cm}$$

Marking guidance: correct substitution; correct square/sum; correct square root.

5. Solution:

$$c^2 = 2.4^2 + 4^2 = 5.76 + 16 = 21.76$$

$$c = \sqrt{21.76} \approx 4.66 \text{ cm}$$

Marking guidance: diagram interpreted correctly; right angle identified; rounding acceptable to two decimal places.

6. Solution: The claim $c = a + b$ is incorrect because Pythagoras' Theorem uses squares, not addition.

$$c^2 = 5^2 + 12^2 = 25 + 144 = 169$$

$$c = 13$$

Marking guidance: clear explanation of misconception; correct working; correct final value.

IMPORTED FROM CHAT:

— [TEACHER ONLY] Consider adding one more representation-shift item (e.g., a rotated triangle) to strengthen transfer.

Copilot Chat:

<https://copilot.microsoft.com/shares/FMduwG7azsTVEF3UnXSeY>

3. [Challenge — Real-World Application] A 5 m ladder leans against a vertical wall. The foot of the ladder sits on level ground, 1.8 m from the base of the wall.

(a) Draw and label a right-angled triangle to represent this situation. Mark the hypotenuse, both known measurements, and the unknown height h .

(b) Calculate h , the height the ladder reaches up the wall. Give your answer correct to one decimal place.

(c) The foot of the ladder is then moved to 2.5 m from the wall. The ladder length stays the same. Without fully recalculating, **explain** whether the ladder now reaches *higher* or *lower* up the wall, and why.

Name: _____

Date: _____

Year 8: Introduction to Pythagoras' Theorem

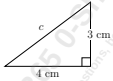
Pythagoras' Theorem applies to right-angled triangles.

$$a^2 + b^2 = c^2$$

where c is the hypotenuse (the longest side).

1. Getting Started

The triangle below is right-angled.



Use Pythagoras' Theorem to find the value of c .

2. Building Confidence

A right-angled triangle has side lengths 5 cm and 12 cm. Find the hypotenuse.

3. Challenge

The hypotenuse of a right-angled triangle is 13 cm. One shorter side is 5 cm. Find the length of the remaining side.

Teacher Answer Key

For teacher use only — not for student distribution

1. Fluency — Finding the hypotenuse (6–8–10 triple)

$$\begin{aligned} c^2 &= 6^2 + 8^2 \\ &= 36 + 64 \\ &= 100 \\ c &= \sqrt{100} = 10 \text{ cm} \end{aligned}$$

Marking: 1 mark — correct substitution into $c^2 = a^2 + b^2$. 1 mark — correct answer $c = 10$ cm.

2. Misconception Check — Finding a shorter side (5–12–13 triple)

(a)

$$\begin{aligned} a^2 &= c^2 - b^2 \\ &= 13^2 - 5^2 \\ &= 169 - 25 \\ &= 144 \\ a &= \sqrt{144} = 12 \text{ cm} \end{aligned}$$

Marking: 1 mark — correct rearrangement $a^2 = c^2 - b^2$. 1 mark — correct answer $a = 12$ cm.

(b) The student **added** the squares ($5^2 + 13^2$) instead of **subtracting** ($13^2 - 5^2$). When finding a shorter side, the theorem must be rearranged to $a^2 = c^2 - b^2$. A further red flag: the student's answer (≈ 13.9 cm) is *larger* than the hypotenuse (13 cm), which is geometrically impossible. Correct answer: $a = 12$ cm.

Marking: 1 mark — correctly identifies addition error. 1 mark — corrected working and answer.

IMPORTED FROM CHAT:

[TEACHER ONLY] Refinement suggestion — add student-facing mark allocations. Currently no marks appear beside each part. For NSW Stage 4 exam readiness, consider annotating each part with (…/2) or similar so students practise gauging how much to write. Q2(b) in particular could become a 2-mark response question with an explicit instruction to give two distinct points (name the error; explain why the answer is impossible). This also doubles as a useful self-assessment prompt.

Claude Chat: <https://claude.ai/share/84170245-13fd-472f-a116-876fc8780987>

Teacher Answer Key

4.

$$c^2 = 4^2 + 3^2$$

$$c^2 = 16 + 9 = 25$$

$$c = 5 \text{ cm}$$

Marking guidance: Substitutes correctly into the formula and calculates $c = 5$ cm.

5.

$$c^2 = 5^2 + 12^2$$

$$c^2 = 25 + 144 = 169$$

$$c = 13 \text{ cm}$$

Marking guidance: Identifies the hypotenuse and applies the theorem correctly.

6.

$$13^2 = 5^2 + b^2$$

$$169 = 25 + b^2$$

$$b^2 = 144$$

$$b = 12 \text{ cm}$$

Marking guidance: Rearranges correctly and solves for the unknown side.

IMPORTED FROM CHAT:

— [TEACHER ONLY] — Add a fourth “spot the error” question where a student incorrectly calculates $3^2 + 4^2 = 7^2$ to explicitly target a common Year 8 misconception about squaring numbers.

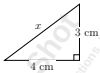
Year 8 Introduction to Pythagoras' Theorem

Key Rule
For a right-angled triangle with shorter sides a and b , and longest side c :

$$a^2 + b^2 = c^2$$

The longest side opposite the right angle is called the hypotenuse.

1. **Starter**
Find the length of the hypotenuse.

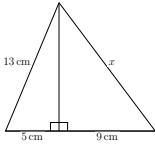


Find x .

2. **Practice**
A right-angled triangle has a hypotenuse of 13 cm and one shorter side of 5 cm.
Find the length of the other side.

3. **Challenge**
A ladder is leaning against a wall. The foot of the ladder is 6 m from the wall and the ladder itself is 10 m long.
How high up the wall does the ladder reach?
Give your answer in metres.

3. **Challenge: The Composite Shape**
The diagram below shows two right-angled triangles joined together. Calculate the length of the side marked x .
Hint: You will need to find the length of the shared middle side first.



IMPORTED FROM CHAT:
I am ready to design your LaTeX worksheet. Could you please provide your teacher brief, including the subject (Maths, Science, or Integrated STEM), Year/Stage, topic, and preferred pedagogy mode?

— [TEACHER ONLY] — *Refinement Suggestion: If you notice students struggling with identifying whether to add or subtract squares, I can add a quick "Misconception Check" diagnostic question before the fluency task where they must spot and explain the error in a provided, flawed calculation.*

Gemini chat:
<https://gemini.google.com/share/4/10EdOeqMA57jSFm7kKp3kr2ZiA5Pa0Lq?usp=sharing>

Teacher Answer Key

- 1.
- $$x^2 = 4^2 + 3^2$$
- $$x^2 = 16 + 9$$
- $$x^2 = 25$$
- $$x = 5$$

Answer: 5 cm
Marking guidance:

- Substitutes correctly into Pythagoras' theorem.
- Squares numbers correctly.
- Gives answer with units.

- 2.
- $$13^2 = 5^2 + b^2$$
- $$169 = 25 + b^2$$
- $$b^2 = 144$$
- $$b = 12$$

Answer: 12 cm
Marking guidance:

- Identifies the hypotenuse.
- Rearranges correctly.
- Gives answer with units.

Teacher Answer Key

- Question 1: Fluency**
- **1a)** Using $c^2 = a^2 + b^2$
 $x^2 = 3^2 + 4^2$
 $x^2 = 9 + 16$
 $x^2 = 25$
 $x = \sqrt{25} = 5$ cm
Marking guide: 1 mark for correct substitution, 1 mark for correct final answer with units.
 - **1b)** Using $a^2 + b^2 = c^2$
 $y^2 + 12^2 = 13^2$
 $y^2 + 144 = 169$
 $y^2 = 169 - 144$
 $y^2 = 25$
 $y = \sqrt{25} = 5$ cm
Marking guide: 1 mark for identifying the hypotenuse is 13 and setting up the subtraction/equation correctly, 1 mark for final answer.

- Question 2: Application**
- **2a)** Students should draw a rectangle with base 2.4 m, height 1.0 m, and a diagonal drawn in. A right-angle symbol should be present in the corner.
 - **2b)** Using $c^2 = a^2 + b^2$
 $c^2 = 2.4^2 + 1.0^2$
 $c^2 = 5.76 + 1.0$
 $c^2 = 6.76$
 $c = \sqrt{6.76} = 2.6$ m
Marking guide: 1 mark for diagram, 1 mark for calculation, 1 mark for final answer in metres.

- Question 3: Challenge**
- **Step 1: Find the shared vertical side (let's call it h)**
Looking at the left triangle:
 $h^2 + 5^2 = 13^2$
 $h^2 + 25 = 169$
 $h^2 = 144$
 $h = \sqrt{144} = 12$ cm
 - **Step 2: Find x**
Looking at the right triangle:
 $x^2 = h^2 + 9^2$
 $x^2 = 12^2 + 81$
 $x^2 = 144 + 81$
 $x^2 = 225$
 $x = \sqrt{225} = 15$ cm
Marking guide: 1 mark for finding the intermediate side (12cm), 1 mark for setting up the second equation, 1 mark for correct final answer.

- 3.
- $$10^2 = 6^2 + h^2$$
- $$100 = 36 + h^2$$
- $$h^2 = 64$$
- $$h = 8$$

Answer: 8 m
Marking guidance:

- Forms the correct equation.
- Solves accurately.
- Includes units.

IMPORTED FROM CHAT:
To draft the worksheet, I need one detail: what is the topic (and year/stage) you want the worksheet to cover? For example: Year 9 Maths - Linear Relationships, Stage 5 Science - Chemical Reactions, or Year 11 Maths Advanced - Trigonometric Functions.

— [TEACHER ONLY] — *Consider adding a misconception-check question where students must identify the hypotenuse before applying Pythagoras' theorem, as Year 8 students commonly square the wrong side.*

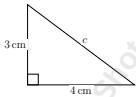
ChatGPT Free: <https://chatgpt.com/share/6a3b1963-6838-83cc-85e5-eebf5448db33>

Introduction to Pythagoras' Theorem

Key idea: In a right-angled triangle, the square of the hypotenuse equals the sum of the squares of the other two sides:

$$a^2 + b^2 = c^2$$

1. **Find the hypotenuse**
A right-angled triangle has side lengths 3 cm and 4 cm.



Find the length of the hypotenuse.

2. **Find a missing side**
A right-angled triangle has hypotenuse 13 cm and one side 5 cm.
Find the length of the other side.

3. **Challenge: Word problem**
A ladder leans against a wall. The base of the ladder is 6 m from the wall, and the ladder is 10 m long.
- How high up the wall does the ladder reach?

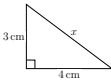
Introduction to Pythagoras' Theorem

Learning Intention

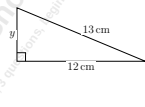
To understand and apply Pythagoras' Theorem ($c^2 = a^2 + b^2$) to find missing side lengths in right-angled triangles. Remember that c must always be the **hypotenuse** (the longest side, directly opposite the right angle).

1. **Fluency: Finding missing sides**
For each of the right-angled triangles below, use Pythagoras' Theorem to calculate the value of the unknown side. Show all steps of your working.

- a) Find the hypotenuse, x .



- b) Find the shorter side, y .



2. **Application: The Farm Gate**
A farmer is building a rectangular wooden gate that is 2.4 m wide and 1.0 m high. To ensure the gate does not sag, she needs to attach a diagonal brace across the back.
- (a) Draw a simple, labelled diagram to represent this scenario.
(b) Calculate the exact length of the diagonal piece of wood required.

Teacher Answer Key

- 1.
- $$c^2 = 3^2 + 4^2 = 9 + 16 = 25$$
- $$c = \sqrt{25} = 5$$
- cm
- Answer:** 5 cm
- 2.
- $$c^2 = a^2 + b^2$$
- $$13^2 = 5^2 + b^2$$
- $$169 = 25 + b^2$$
- $$b^2 = 144$$
- $$b = \sqrt{144} = 12$$
- cm
- Answer:** 12 cm

- 3.
- $$c^2 = a^2 + b^2$$
- $$10^2 = 6^2 + h^2$$
- $$100 = 36 + h^2$$
- $$h^2 = 64$$
- $$h = \sqrt{64} = 8$$
- m
- Answer:** 8 m

IMPORTED FROM CHAT:
I'm ready to build this, but I need a bit more detail first. What are the key inputs: subject (Maths/Science), year or stage, topic, number of questions, and desired difficulty or mode (e.g. exam-style, scaffolded, investigation)?

— [TEACHER ONLY] —
Add a misconception check (e.g. a triangle that is not right-angled) to ensure students do not apply Pythagoras indiscriminately.

Perplexity Chat:
<https://www.perplexity.ai/search/90c8d150-9963-463f-8268-ee5ea7ac3962>

Name:

Date:

My Measurement Challenge!

Learning to measure with cubes and rulers

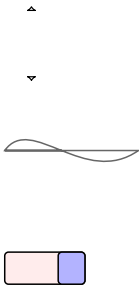
Part 1: Match the Words

Draw a line from the word to the matching picture.

String

Rubber

Tall



1. **Word Bank Challenge:** Use the words below to finish the sentence.

string tall rubbers

To find out how my tower is, I can measure it with .

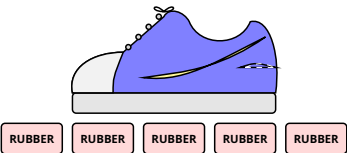
Part 2: Hands-On Non-Standard Units

Grab your measuring tools and let's get to work!

2. **Measure Your Desk!**
Place your hands flat on your desk, side-by-side. Measure how long your desk is from one side to the other.

My desk is hands long.

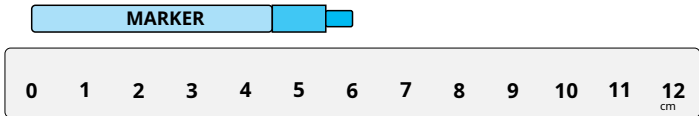
3. How many rubbers long is this shoe?



The shoe is rubbers long.

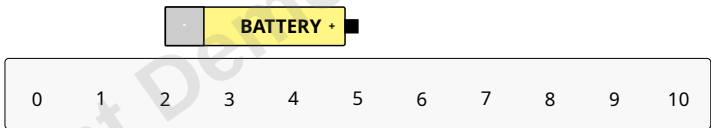
Part 3: Using a Ruler (cm)

4. Use the ruler below to measure the marker in centimetres (cm).



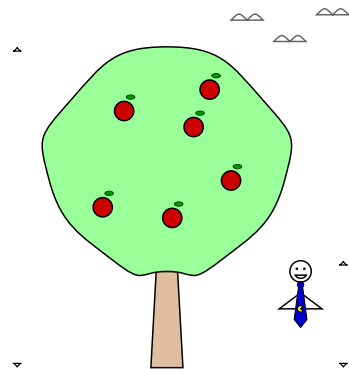
The marker is cm long.

5. **Level 2:** This battery does **not** start at 0. Find the length.



Starts at: Ends at: Length: cm.

6. **Tree Challenge:** Measure the **height** of the tree and Mr Darra.



The tree drawing is cm tall.

Mr Darra is cm tall.

Part 4: Length Logic

7. **Ordering (Baseline):** Order these lengths from **shortest** to **longest**.

18 cm, 64 cm, 42 cm, 110 cm

Shortest: → → → Longest:

8. **Estimation:** How tall is Ms Circle one.

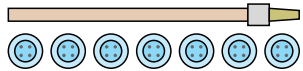
170 cm 17 cm 17 m

Differentiated Challenges

Extra tasks for measuring with blocks and rulers!

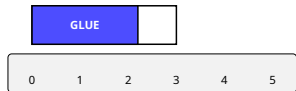
Level 1: Foundation (Easier)

9. How many **buttons** long is the paintbrush?



The paintbrush is buttons long.

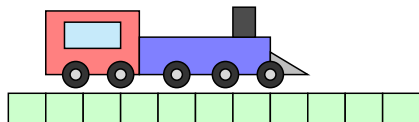
10. Measure the **Glue Stick** with the ruler below.



The glue stick is cm.

11. The glue stick is (**shorter** / **longer**) than the paintbrush.

12. How many **blocks** long is the toy train?

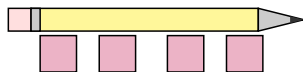


The toy train is blocks long.

13. Which word is a **unit** of length? Circle the correct answer.

Triangle Metre Green

14. Mr Darra measured this pencil with blocks. Did he do it correctly? Circle YES or NO.



YES NO

Level 2: Intermediate (Medium)

15. Find the total length if you put the marker and the battery together.

Hint: Marker = 6 cm, Battery = 3 cm. What is the total?

Working: Total: cm.

16. Write a word from your word bank that can be used to measure a curved line.

Answer:

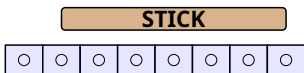
Level 3: Extension & Reflection

17. **Super Challenge (Mixed Units):** Order these lengths from shortest to longest.

15 cm, 3 m, 110 cm, 1 m, 80 cm

Shortest: → → → → Longest:

18. This stick does **not** start at the first cube. How many cubes long is it?



The stick is cubes long.

19. Write a sentence explaining **how** you found the length of the stick when it didn't start at the beginning.

20. **Word Problem Challenge:** A piece of wood is 1 metre long. Mr Darra cuts off 20 cm. How long is the wood now?

Working: Answer: cm

21. **The Measurement Trap!** Mr Darra asked two students to measure items.

-  measured a Pencil Case. It was **2 pencils** long.
-  measured a Glue Stick. It was **3 rubbers** long.



 says: "3 is bigger than 2, so my Glue Stick must be longer!"

Explain why we cannot measure one thing with rubbers and another thing with pencils to compare their sizes:

Reading for Fluency

Let's practice reading like experts!

The 3 Keys to Good Reading

1. **Accuracy:** We sound out tricky words instead of guessing.
2. **Expression:** We don't read like a robot! Our voice goes up at a question mark (?) and we take a breath at a full stop (.).
3. **Understanding:** We make a movie in our head about what the words mean.

22. **Paired Reading:** Read this story to your partner using the 3 Keys.

Today, Mr Darra showed us how to measure. We learned that we must use the same unit to compare sizes! If  uses pencils and  uses rubbers, we get confused.  raised his hand and asked, "But what if my pencils are short?" Mr Darra smiled. "That is a great question!" he said. Then,  asked, "Can we just use a ruler instead?" We all agreed that rulers make measuring fair. Do you know how to use a ruler?

23. **My Reading Checklist:** Tick the boxes after you read!

- ☐ I sounded out the tricky words.
- ☐ I stopped at the full stops.
- ☐ I know what the sentence means.



24. **Writing: Teach an Alien!**

Write 3 sentences to teach an alien how to measure your desk.

First, I

Next, I

Finally, I

25. **Reflection:** What was the trickiest part about measuring today?

I thought

was tricky because

